PREM KUMAR S

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PROFESSIONAL SUMMARY

Motivated Artificial Engineering and machine learning undergraduate with a strong foundation in programming, cloud computing, and artificial intelligence. Proficient in Python, C, and HTML, with hands-on experience in developing AI-based applications. Certified in AWS Cloud Foundations and AWS Cloud practitioner Essential, with a keen interest in building scalable, real-world solutions.

EDUCATIONAL QUALIFICATION

Senior Secondary School
04/2023
Rajavignesh Higher secondary school preambular, India
Percentage: 85%

Bachelor of Technology (B.Tech), Artificial intelligence and machine learning
2023 – present
Saveetha Engineering College Chennai
• Pursuing
• CGPA: 8.00

PROJECTS COMPLETED

Satellite Image Detection Using Mathematical CLIP Model
08/2025 – 10/2025
• To automatically detect and classify objects or land areas from satellite images.
• Convert images into feature vectors

Creative Web Poster Showcase
• To design attractive and responsive digital posters using HTML and CSS.
• To improve frontend development and UI/UX design skills.

Research Paper Analysis System using Retrieval-Augmented Generation (RAG)
• Developed an intelligent document retrieval and analysis system using RAG architecture that automatically extracts, summarizes, and contextualizes key insights from research papers, reducing manual literature review time by 60%.
• Implemented semantic search and vector-based document indexing capabilities enabling users to query research papers in natural language and receive contextually relevant answers from multi-document sources.

Universal LLM-Powered Chatbot Integration Platform
• Engineered a multi-application chatbot framework leveraging large language models that seamlessly integrates with 5+ platforms (Slack, Teams, Discord, web, mobile) enabling unified conversational AI access across enterprise applications through a single backend.
• Designed intelligent context-switching and application-specific prompt engineering system allowing the chatbot to adapt responses based on platform requirements and user intent, improving response accuracy by 45% and reducing integration time for new applications by 70%.

Integrated Medical Image Diagnostics & Chatbot Platform with Report Generation
• Developed an AI-powered medical imaging platform that combines deep learning-based image analysis, an intelligent healthcare chatbot, and automated PDF report generation. The system analyzes X-rays, CT scans, and MRI images with 92% accuracy, provides real-time diagnostic insights via conversational AI, generates comprehensive clinical reports, and enables instant download of findings.
• Engineered end-to-end diagnostic pipeline integrating deep learning image analysis with chatbot interface that automatically processes medical images, extracts clinical findings, answers patient queries in real-time, and

generates professional PDF reports with findings, recommendations, and clinical summaries—reducing reporting time by 70%.

Privacy-First Local AI Text-to-SQL Pipeline for Enterprise Data

• Architected privacy-first Text-to-SQL pipeline using local Ollama LLM that converts natural language questions into SQL queries without external cloud dependencies, ensuring 100% data sovereignty and eliminating security vulnerabilities associated with API-based solutions.

• Implemented air-gapped, on-premise infrastructure with zero external API dependency that processes database queries in milliseconds while maintaining compliance with data residency requirements, reducing query processing time by 80% and eliminating cloud costs.

CERTIFICATIONS

AWS Academy Graduate - Cloud Foundations	Foundations and Core Concepts of PyTorch
August 2025	September 2025
AWS Cloud Practitioner Essentials	LinkedIn Learning - Learning Frontend Development
November 2025	December 2025

TECHNICAL SKILLS AND SOFT SKILLS

Technical Skills	**Soft Skills**
PYTHON | C | HTML | MACHINE LEARNING | AWS CLOUD FUNDAMENTAL | JAVA | JAVASCRIPT | GIT HUB	Time management | Flexible and adaptable | Calm under pressure | Active listening | Multitasking Abilities